Real-unit-calibrated simulations close the sim-to-real gap for CNN Einstein-radius regression, with self-diagnosed confidence

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Abstract

- Task: CNN predicts Einstein radius θ_E from a single HST ACS/WFC F814W lensed image.
- Prior CNN trained on simplified simulations: good in-distribution, fails on real lenses ($R^2 < 0$, prior-pull).
- Fix: paltas/lenstronomy + real COSMOS sources + real STScI focus-diverse ePSF + real empty-cutout backdrops + noise calibrated to the real SLACS sky-RMS distribution ("hybrid" training images).
- Result: R^2 crosses negative to positive on 62 SLACS + 40 S4TM (frozen, never trained on).
- New addition: $(\mu, \log \sigma^2)$ uncertainty head — σ correlates with real error ($\rho = +0.71$, SLACS); confidence-gated subset meets the target success bar outright.
- Positioning vs. Cao et al. 2025 [cite] (same lenses, same b_{SIE} ground truth, conventional pixel-based modeling, ~3 min/lens): CNN matches on the confident subset at roughly 10^5 times the inference speed.

1. Introduction

- Strong-lensing θ_E is a standard mass-scale probe; upcoming surveys (LSST, Euclid, Roman) are expected to discover order 10^5 lenses — amortized CNN inference is the only tractable option at that scale.
- Prior CNN work:
  - Hezaveh, Perreault Levasseur & Marshall 2017, *Nature* [cite] — first CNN regression of SIE parameters from HST-like simulations.
  - Perreault Levasseur, Hezaveh & Wechsler 2017, *ApJL* [cite] — CNN uncertainty estimation for lens parameters.
  - Gawade et al. 2025, arXiv:2404.18897 — ground-based HSC CNN, ~10-20% fractional error.
  - [cite, verify not duplicate of above] arXiv:1911.06341 — gri wide-field CNN, θ_E to ~10-15%.
  - Cao et al. 2025, arXiv:2503.08586 [cite] — not a CNN; automated pixel-based modeling + nested sampling on 63 grade-A SLACS lenses, $\lesssim 5\%$ deviation, ~10% catastrophic failure rate, ~3 min/lens. Note their own discussion flags ML priors as a path to reducing failures.

- Open problem this paper addresses: CNNs trained on simplified simulations show a sim-to-real generalization gap on real telescope data.
- Contributions:
  1. Diagnose the prior (pre-paltas) model's real-lens failure as **prior-pull** — systemic regression toward the training-set mean, not an architectural defect.
  2. A physically-calibrated "hybrid" simulator: paltas/lenstronomy + real COSMOS sources + real focus-diverse ePSF (STScI ISR 2018-08 / ISR 2023-06 [cite]) + real empty-field backdrops + noise matched to the real sky-RMS distribution — no hand-tuned flux normalization.
  3. R^2 crosses negative to positive on the frozen real benchmark; the residual gap is quantified via the prediction-vs-truth regression slope.
  4. An uncertainty head whose σ is domain-aware and predictive of real-world failure; gating on it isolates a subset that meets the target success bar.

2. Data

2.1 Real benchmark (frozen, never used in training)

- $N = 62$ SLACS + 40 S4TM [cite S4TM], HST ACS/WFC F814W.
- Ground truth: Bolton et al. 2008 SIE b_{SIE} [cite].
- One SLACS lens (J0955+0101) excluded — corrupted cutout.
- Cutouts: 128x128 px at 0.05″/px (6.4″ field of view), MAST drizzled products.

2.2 Training simulator ("hybrid")

- Base: paltas [cite] on lenstronomy; PEMD+shear deflector; COSMOS_23.5 source catalog (real, GREAT3-vetted galaxies).
- $\theta_E \sim U[0.45, 2.30]$″, flat.
- Lens light: joint empirical (magnitude, R_{Sersic}) prior, drawn from 63 (magnitude, R_e) pairs measured directly from the real SLACS cutouts, with jitter.
- PSF: STScI focus-diverse empirical PSFs (real HST optics), 88 distinct kernels from real archival exposures, randomly rotated per generation shard.
- Backdrop: noiseless simulated render + a real empty COSMOS cutout (real correlated noise and field neighbors) + Gaussian top-up noise drawn per-image from the real SLACS sky-RMS distribution.

Table 1. Dataset realism gate (sim vs. real, identical estimator both sides).

Metric	Simulated	Real	Target
Sky RMS ratio (sim/real)	1.14	—	0.8 - 1.25
Lens peak / sky	723	492 - 1798 (16-84%)	within real band
θ_E range	[0.45, 2.30]	[0.54, 1.78]	benchmark inside interior

- 100,000 training / 5,000 validation images; validation uses seed- and PSF-kernel-disjoint generation.

Figure 1. Real vs. simulated cutouts under an identical stretch. *[insert figure]*

2.3 Current limitations of the simulator

- Lens light is a parametric Sersic profile, not a real elliptical galaxy image (real deflectors show disky/inclined morphology, dust, tidal features).
- Empty-field backdrops drawn from a single COSMOS tile (1,317 unique cutouts, reused across the training set); a larger backdrop pool is in preparation.
- Mass model is SIE/PEMD only; no hydrodynamical-simulation mass structure yet (see Section 6).

3. Model and training

- Architecture: scale-conditioned ResNet (four residual stages, global average pooling, scale-scalar concatenation, MLP head).
- Input: single-band, asinh-normalized image (per-image mean/std after arcsinh compression).
- Loss: point estimate (Huber) in an initial version; $(\mu, \log\sigma^2)$ Gaussian negative log-likelihood in the reported model (uncertainty head).
- Optimizer: Adam, cosine learning-rate schedule, 40 epochs, batch size 64; flip/rotation augmentation (θ_E is invariant under all eight dihedral transforms).
- Model selection: simulation validation split only; the real benchmark is evaluated at fixed checkpoints, never used for selection.
- Test-time augmentation: prediction averaged over the eight dihedral views; total predicted uncertainty combines the mean aleatoric variance with the across-view (epistemic) variance.

4. Results

4.1 Sim-to-real failure diagnosis (baseline, pre-hybrid-pipeline)

Table 2. Baseline model on the frozen real benchmark.

Sample	N	Median frac. error	R^2	MAE	Failure rate ($> 15\%$)
SLACS	62	-7.9%	-1.02	0.274″	55%
S4TM	40	-5.3%	-0.53	—	68%

- Diagnosis: signed error sweeps from positive to negative across true θ_E, crossing zero near a training-mean attractor (~0.95″-1.0″) — prior-pull. Weak per-feature correlation with the residuals (all $|\rho| < 0.3$) indicates a systemic effect rather than an architectural or image-specific one.

4.2 Hybrid pipeline, iteration 1

Table 3. Hybrid model, iteration 1, on the frozen real benchmark (bootstrap 95% CI, 10,000 resamples).

Sample	N	Median frac. error (95% CI)	R^2	MAE	Failure rate (95% CI)
SLACS	62	-1.1% [-4.0, $+2.2$]	$+0.23$	0.139″	27% [18, 39]
S4TM	40	$+5.8\%$ [$+2.1$, $+16.9$]	$+0.33$	0.159″	38% [22, 52]

- R^2 crosses from negative to positive on both samples relative to the baseline.
- The residual S4TM bias traces to lenses below the training prior's lower bound (true θ_E down to $0.54″ against a training floor of $0.6″).

4.3 Hybrid pipeline, iteration 2 (widened prior, expanded PSF diversity, uncertainty head, test-time augmentation)

Table 4. Hybrid model, iteration 2, on the frozen real benchmark.

Sample	N	Median frac. error (95% CI)	R^2	MAE	Failure rate (95% CI)	Slope
SLACS	62	-2.6% [-4.5, -0.1]	$+0.27$	0.129″	23% [13, 34]	0.72
S4TM	40	-1.8% [-7.9, $+1.5$]	$+0.47$	0.140″	38% [22, 52]	0.71

- Both medians now fall within a $\pm 5\%$ target band; the prediction-vs-truth regression slope (1.0 = no compression) improves from 0.62 to 0.72 (SLACS) and 0.58 to 0.71 (S4TM), quantifying a reduced but still-present domain-gap compression.

Figure 2. Predicted vs. true θ_E: baseline model, hybrid iteration 2, and the Cao et al. 2025 accuracy band. *[insert figure]*

4.4 Confidence-gated performance

- Spearman correlation between predicted σ/μ and true absolute fractional error: $+0.71$ (SLACS), $+0.39$ (S4TM) — the model's own uncertainty estimate is predictive of real-world error, not only of simulation-validation error.
- σ/μ is itself domain-aware: a median of roughly 9-10% on real lenses versus roughly 1% on simulation validation, without any explicit domain label at training time.

Table 5. Confidence-gated subsets.

Sample	Subset	N	Median frac. error	Failure rate
SLACS	full	62	-2.6%	23%
SLACS	$\sigma/\mu \leq$ median	31	-1.6%	**6%**
SLACS	$\sigma/\mu \leq$ 75th pct.	46	-1.5%	7%
S4TM	full	40	-1.8%	38%
S4TM	$\sigma/\mu \leq$ median	20	-0.9%	15%

- The confident half of the SLACS sample meets the target success bar outright ($\pm 5\%$ median, $\lesssim 10\%$ failure) and is competitive with the Cao et al. 2025 failure rate at substantially higher inference speed.

4.5 Planned but not yet completed

- Ablation: flat vs. skewed θ_E prior.
- Ablation: benchmark-matched vs. broad (non-benchmark) PSF pool.
- Ablation: real backdrop vs. Gaussian noise only.
- Ablation: real empirical PSF vs. Gaussian/Moffat PSF.

- Matched per-lens comparison against Cao et al. 2025's public per-lens results.
- S4TM-specific noise recalibration (current noise model calibrated to SLACS sky-RMS only).
- Multi-parameter (ellipticity, centroid) re-validation on the hybrid pipeline.

5. Discussion

- The crossing of R^2 from negative to positive is the central result: the baseline model did not track individual real lenses; the hybrid-trained model does.
- The residual gap is compression (slope < 1) rather than bias (median ≈ 0): the model hedges toward the prior mean under domain shift, a milder form of the baseline's prior-pull.
- The uncertainty head serves three purposes: a calibration/error-bar deliverable, a practical deployment filter, and an ingredient (mean-variance estimation) that pairs naturally with unsupervised domain adaptation (Section 6.2).
- Standing limitation: the lens light remains parametric. If a fully realism-calibrated model still showed $R^2 < 0$ on real lenses, that would motivate escalating to real deflector-cutout injection; this has not been triggered, as R^2 is positive.

6. Proposed extensions

6.1 IllustrisTNG convergence maps within paltas

- Replace or augment the SIE/PEMD deflector with deflection fields derived from IllustrisTNG convergence maps, introducing real (non-elliptical) mass structure: twists, substructure, boxiness.
- Two risks require checking before use: (i) a label-convention offset between SIE b_{SIE} and the mean-convergence-based θ_E definition, to be quantified on a TNG subsample before mixing label types; (ii) the known IllustrisTNG central-image artifact from limited hydrodynamical resolution [Bolton 2012; Shu et al. 2016 — cite], a new sim-to-real gap that this change would introduce and that must be checked for.

6.2 Sim-to-real domain adaptation

- Prior domain-adaptation work for lens θ_E regression exists but adapts between two simulated domains, not from simulation to real data: Ciprijanovic et al. [cite, arXiv:2311.17238] (adversarial and maximum-mean-discrepancy adaptation) and [cite, arXiv:2411.03334] (unsupervised domain adaptation combined with a mean-variance estimator).
- This leaves open a sim-to-real domain adaptation validated against a real spectroscopic benchmark, which this project's frozen 62+40 sample with b_{SIE} ground truth is positioned to provide.
- Any unsupervised adaptation must draw its unlabeled target pool from real lenses disjoint from the frozen benchmark, to avoid contaminating the evaluation set.
- Sequenced after the ablations in Section 4.5, so that the pre-adaptation sim-to-real gap is fully characterized first.

7. Conclusion

- [to be written after ablations are complete]

References

- Bolton et al. 2008, SLACS b_{SIE} catalog [cite]
- Hezaveh, Perreault Levasseur & Marshall 2017, *Nature* 548, 555 [cite]
- Perreault Levasseur, Hezaveh & Wechsler 2017, *ApJL* [cite]
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- Ciprijanovic et al., arXiv:2311.17238
- [cite] arXiv:2411.03334
- [cite] arXiv:2410.16347
- Bolton 2012; Shu et al. 2016 [cite]
- Wan, Chan, Lange & Hannuksela, LensFusion [cite]